Conditional statements

1

a No **b** Yes **c** No **d** No

e No **f** No **g** Yes **h** No

i No **j** No **k** Yes **l** No

m No **n** Yes **o** No **p** Yes

q Yes **r** No **s** No **t** No

u No **v** No **w** Yes **x** Yes

2

a No **b** No **c** Yes **d** Yes

e No **f** No **g** Yes

3

a If two lines are parallel, then they never intersect.

b If the animal is a cat, then it cannot speak French.

4

a Yes **b** No

5

a Some birds are not able to fly.

b All turtles have claws.

c Some Libras are Geminis.

d Some bicycles have four wheels.

e Some houses are not connected to the internet.

6

a If D, then C. **b** If $\sim C$, then $\sim D$.

c If $\sim D$, then $\sim C$.

7

a If an animal has claws, then it is a bear.

b If a boomlug is delicious, then it is round.

c If a figure is a quadrilateral, then it is a square.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8

a If an animal does not have feathers, then it is not a bird.

b If a number is not divisible by ten, then it does not end in zero.

c If a bamboozle is not loud, then it is not square.

9

a If an animal is not a bird, then it does not have feathers.

b If two angles do not form a right angle, then they are not complememtary.

c If a heffelump is not lost in the woods, then it is not a woozle.

10 Converse

Additional questions

11 If p, then q.

12 No, a conditional statement does not need to be true.

13 The following statements are conditional statements:

a If a polygon has three sides, then it is a triangle.

c If I go to school in Florida, then I live in Orlando.

d If it is raining, then we cannot play baseball.

e If a parallelogram is not a rhombus, then it is not a square.

14

a Hypothesis: Two angles are vertical
Conclusion: They are congruent

b Hypothesis: The sum of a polygon's interior angle measure sum is 180°
Conclusion: The polygon is a triangle

c Hypothesis: Two triangles have the same side lengths
Conclusion: They are congruent

d Hypothesis: Two lines are parallel
Conclusion: The consecutive interior angles formed by a transversal are supplementary

15

a If I am an artist, then I can paint a picture.

b If I cannot paint a picture, then I am not an artist.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c If I am not an artist, then I cannot paint a picture.

16

a Converse statements

b Contrapositive statements

c Inverse statements

17

a False. I can go to any store or online retail website to go shopping. For example, I can go to Southeastern Grocers if I need to go shopping.

b True

c True

d False. Many different people go to schools for work or to visit. For example, if I go to school I could be a teacher.

18

a Yes, this statement is true.

b No. The converse statement, "if $x^2 = 4$, then $x = 2$" is not true because $x = -2$ is also possible.

c No. The inverse statement, "if $x \neq 2$, then $x^2 \neq 4$" is not true because $x = -2$ is possible.

d Yes, the contrapositive is true.

19 If a polygon is a square, then it has four right angles.
The converse of this statement "If a polygon has four right angles, then it is a square" is not true because a rectangle has four right angles and is not necessarily a square.

20

a If two angles are vertical, then they are congruent.

b If two angles are not congruent, then they cannot be vertical.

21

a If you don't take a shot, then you can never score it.

b If they are bigger, then they will fall harder.

c If you are in doubt, then tell the truth.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647